CleanGoa

Together, Let’s Keep Goa Clean

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A project Report For

CSA 5005- Object Oriented Technology Lab

Submitted By

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ABSTRACT

CleanGoa is a web-based platform developed to improve cleanliness and protect the environment in Goa by connecting citizens, volunteers, and local administrators. The main aim of this project is to provide a simple and effective way for people to report cleanliness issues and participate in activities that help maintain clean and healthy surroundings.

The website allows users to report problems such as garbage dumping and waste accumulation by uploading photographs, providing a description, and selecting or entering the location of the issue. Administrators can review the submitted reports and approve or reject them based on the information provided. Approved reports are displayed on a public map, allowing users to view cleanliness issues in different locations and track their current status. This helps improve awareness and makes it easier to identify areas that require attention.

CleanGoa also includes a Volunteer Mission Management module that allows administrators to organize cleanliness drives in public places, including beaches and other common areas. Users can view mission details, register for upcoming activities, and participate in community cleaning efforts. This feature encourages teamwork and active public involvement in maintaining cleanliness.

The project aims to improve communication between citizens and local administrators, encourage responsible waste disposal, and promote environmental awareness. By bringing cleanliness reporting, public issue tracking, and volunteer mission management together on a single platform, CleanGoa seeks to make cleanliness-related activities more organized, accessible, and efficient. Overall, the project supports the vision of a cleaner and more environmentally responsible Goa through the combined efforts of the community.

CHAPTER 1

INTRODUCTION

1. INTRODUCTION

The development of CleanGoa is aimed at improving cleanliness and protecting the environment in Goa through a web-based platform. Cleanliness is an important concern, especially in tourist destinations, beaches, public places, and residential areas where garbage and improper waste disposal can affect the environment and public health. To address these issues, CleanGoa provides a common platform where citizens, volunteers, and local administrators can work together to maintain cleaner surroundings.

The Cleanliness Reporting module allows users to report problems such as garbage dumping and waste accumulation by uploading photographs, providing a description, and specifying the location of the issue. Administrators can review the submitted reports and approve or reject them. Approved reports are displayed on a public map, allowing users to view cleanliness issues and track their current status.

The Volunteer Mission Management module focuses on encouraging people to participate in cleanliness activities. Administrators can organize cleanliness drives at beaches, public areas, and other locations. Users can view the mission details and register to participate in these activities. This helps promote teamwork and increases public awareness about environmental protection.

The development of CleanGoa aims to make cleanliness reporting easier, improve communication between citizens and administrators, and encourage community participation. By combining issue reporting, public mapping, and volunteer management in one website, CleanGoa seeks to support better cleanliness management and contribute towards a cleaner and greener Goa.

1.1 BACKGROUND

The development of CleanGoa is motivated by the need to improve cleanliness and waste management in Goa through the use of digital technology. Goa is known for its beautiful beaches, tourist destinations, and natural environment. However, problems such as garbage accumulation, littering, and improper waste disposal can affect the cleanliness of public

places and harm the surrounding environment. Addressing these issues requires proper reporting, effective coordination, and active participation from the public.

In many situations, citizens may find it difficult to report cleanliness issues, share accurate location details, or track the progress of a complaint. Similarly, local administrators need an organized way to review reported problems and manage cleanliness-related activities. There is also a need to encourage volunteers to participate in cleaning drives and contribute to environmental protection.

CleanGoa is proposed as a web-based platform that brings these activities together in one system. The website allows users to report cleanliness issues by uploading photographs, providing descriptions, and specifying the location of the problem. Administrators can review the submitted reports and approve or reject them. Approved reports can be displayed on a public map, allowing users to view reported issues and track their current status. The platform also includes a Volunteer Mission Management module that enables administrators to organize cleanliness drives and allows users to view mission details and register for participation. This helps improve coordination and encourages people to work together to maintain cleaner surroundings.

The project focuses on the following areas:

- Digital cleanliness reporting**:** Providing a convenient way for citizens to report garbage and other cleanliness problems.
- Location-based visualization: Displaying approved reports on a public map to help users identify affected areas.
- Administrative control: Allowing administrators to review reports and manage cleanliness-related activities.
- Volunteer coordination: Supporting the organization of cleanliness drives and participation by interested users.

Overall, CleanGoa provide a simple and organized digital solution for cleanliness reporting and volunteer coordination. By connecting citizens, volunteers, and local administrators through a common platform, the project seeks to improve communication, support cleanliness management, and contribute towards a cleaner and greener Goa.

1.2 AIM AND OBJECTIVES

The aim of CleanGoa is to develop a user-friendly web-based platform that connects citizens, volunteers, and local administrators to improve cleanliness and protect the environment in Goa. The project focuses on simplifying the reporting of cleanliness issues through photographs, descriptions, and location details. It enables administrators to review reports and display approved issues on a public map for easy tracking. The platform also supports the organization of cleanliness drives and allows users to register as volunteers. Through these features, CleanGoa aims to improve coordination, encourage community participation, promote responsible waste management, and contribute towards a cleaner Goa.

To achieve the primary aim of cleanliness, the project focuses on following objectives-

- To develop a centralized platform: Create a single website for cleanliness reporting and volunteer mission management.
- To simplify issue reporting: Allow users to report garbage dumping and other cleanliness issues by uploading photographs, adding descriptions, and providing location details.
- To implement report management: Enable administrators to review submitted reports and approve or reject them.
- To provide a public map: Display approved cleanliness reports on a map so users can identify affected areas.
- To enable status tracking: Allow users to check the current status of their submitted reports.
- To manage volunteer missions: Provide features for administrators to organize cleanliness drives by specifying the location, date, time, and other relevant details.
- To improve communication and coordination: Connect citizens, volunteers, and administrators through a common digital platform.
- To promote environmental awareness: Encourage responsible waste disposal and active participation in maintaining clean public spaces.
- To support a cleaner Goa: Use digital technology to make cleanliness-related activities more organized and accessible, contributing to environmental protection.

1.3 PROBLEM STATEMENT

Cleanliness and waste management are important challenges in maintaining a healthy environment and protecting the natural beauty of Goa. Public places, beaches, tourist destinations, and residential areas may experience problems such as garbage accumulation, littering, and improper waste disposal. These issues can negatively affect the environment, public health, and the overall appearance of the surroundings.

One of the main problems is the difficulty citizens may face in reporting cleanliness issues and communicating the exact location and details of the problem to the concerned authorities. Without a centralized platform, it can be challenging to maintain organized records of reported issues, monitor their progress, and provide updates to the people who reported them. Citizens may also find it difficult to identify existing cleanliness problems in nearby areas.

Another challenge is the coordination of cleanliness activities and volunteers. Organizing cleaning drives requires proper communication, sharing of information, and coordination among interested participants. Without a common platform, information about upcoming cleanliness activities may not reach enough people, which can limit public participation.

Therefore, there is a need for an integrated digital system that simplifies cleanliness reporting, improves the visibility of reported issues, supports administrative review, and encourages people to participate in environmental protection activities. CleanGoa is proposed to address these challenges through a single web-based platform that connects citizens, volunteers, and local administrators.

1.3 CURRENT GAPS AND PROPOSED SOLUTION

1.3.1 CURRENT GAPS

Existing methods of reporting and managing cleanliness issues may involve separate communication channels, manual processes, or limited access to information. These methods may not provide a single place for citizens to report problems, view their locations, track their status, and participate in cleanliness activities.

The major gaps identified are:

- Lack of a centralized reporting system: Citizens may need to use different methods to communicate cleanliness problems, making information difficult to organize and manage.
- Difficulty in identifying exact locations: Reports without clear photographs or location details can make it harder to understand where a cleanliness issue has occurred.
- Limited visibility of reported issues: Citizens may not have convenient access to a common map showing reported cleanliness problems in different areas.
- Lack of transparent status tracking: People may find it difficult to know whether a report has been received, reviewed, approved, rejected, or resolved.
- Challenges in administrative management: Reviewing reports and maintaining records can become difficult when information is not organized in a centralized system.
- Limited volunteer coordination: Information about cleanliness drives, including their locations, dates, and timings, may not reach all interested participants.
- Lack of an integrated platform: Cleanliness reporting and volunteer activities may be managed separately, making coordination between citizens, administrators, and volunteers more difficult.
- Need for greater public participation: People may be willing to contribute to cleanliness activities but may lack an accessible platform to report issues or find opportunities to volunteer.

These gaps show why we really need a simple, easy-to-use app or website to report trash and get the community involved. That is the exact challenge CleanGoa is built to solve.

CleanGoa fixes this by bringing everyday citizens, volunteers, and local authorities together onto one simple platform. By doing this, it helps everyone work hand-in-hand to build a cleaner, greener, and more accountable community.

1.3.2 PROPOSED SOLUTION

CleanGoa proposes a web-based platform that brings cleanliness reporting, administrative review, public issue tracking, and volunteer mission management together in one system. The website is designed to make it easier for citizens to report problems, for administrators to manage submissions, and for volunteers to participate in organized cleanliness activities.

The proposed solution includes the following features-

- Online cleanliness reporting: Users can submit reports about garbage dumping, littering, and other cleanliness issues by uploading photographs, entering descriptions, and providing location details.
- Location-based reporting: Users can select a location on a map or enter it manually, helping administrators identify the affected area.
- Administrative review and approval: Administrators can examine submitted reports and approve or reject them based on the information provided.
- Public cleanliness map: Approved reports can be displayed on a public map so users can view reported issues and understand their geographical distribution.
- Report status tracking: Users can check the current status of their reports, improving visibility into the reporting process.
- Volunteer mission management: Administrators can create cleanliness missions by providing details such as the activity description, location, date, and time.
- Volunteer registration: Users can browse available missions, view the relevant details, and register to participate in cleanliness drives.
- Centralized information management: The platform can maintain organized records of users, cleanliness reports, report statuses, and volunteer missions.
- Improved communication and coordination: By bringing citizens, volunteers, and administrators together, CleanGoa provides a common platform for sharing information and coordinating cleanliness-related activities.
- Environmental awareness: The website encourages responsible waste disposal, community involvement, and collective efforts to maintain cleaner public spaces.

CleanGoa bridges the gap between citizens, volunteers, and local authorities by creating a transparent, all-in-one platform for reporting issues, tracking progress, and organizing community cleanups.

1.3.3 HOW CleanGoa ADDRESSES THE CURRENT GAPS

CleanGoa solves the everyday hassles of reporting trash and organizing cleanups by putting everything onto one simple, easy-to-use website. It brings everyday people, volunteers, and local leaders together so everyone can team up to protect the environment and keep our neighbourhoods clean.

Easy Reporting and Tracking

The website makes reporting cleanliness issues quick and simple:

- Submit Reports: Anyone can easily snap a photo, write a quick description, and pinpoint where the trash or littering is located.
- Review & Mapping: Local administrators check these reports and approve them. Once approved, the issues show up on a public map so everyone can see where the problems are.
- Track Progress: Users can check on their reports anytime to see if they are being looked into or fixed, making the whole process completely transparent.

Bringing Volunteers Together

CleanGoa also makes it much easier to organize community cleanups through its Volunteer Mission Management section:

- Organize Drives: Administrators can easily set up cleanup events by adding the date, time, location, and details of what needs to be done.
- Join In: People can browse these missions and sign up with just a few clicks. It gives everyone a fun, organized way to pitch in and help out.

A Greener Future for Goa

By combining reporting, map views, admin reviews, and volunteer sign-ups into one single place, CleanGoa cuts through the confusion and gets things done faster. Looking ahead, the platform has room to grow with helpful new features like instant notifications and data reports. Overall, CleanGoa is here to spread environmental awareness, encourage better waste disposal, build a stronger community spirit, and help create a cleaner, greener, and more sustainable Goa for everyone.

1.4 SCOPE

The primary scope of CleanGoa is to develop a web-based platform for cleanliness issue reporting and volunteer mission management, with the aim of improving cleanliness and promoting environmental protection in Goa. The project focuses on providing a centralized system that connects citizens, volunteers, and local administrators to support the identification and management of cleanliness issues.

The platform allows users to report problems such as garbage accumulation, littering, and improper waste disposal by uploading photographs, providing descriptions, and specifying locations. Administrators can review the submitted reports, approve or reject them, and manage their status. Approved reports are displayed on a public map, enabling users to view reported issues and track their current status. The project also includes a Volunteer Mission Management module through which administrators can organize cleanliness drives by specifying details such as location, date, time, and activity description. Users can view available missions and register to participate. These features aim to improve coordination, encourage community involvement, and promote responsible waste management.

The scope further includes the organized management of information related to users, cleanliness reports, report statuses, and volunteer missions. However, the actual resolution of reported issues depends on the involvement of responsible authorities and participating communities.

The final deliverable of the project is a functional web-based CleanGoa platform that integrates cleanliness reporting, administrative review, public map visualization, report status tracking, and volunteer mission management. The completed system aims to provide a convenient and organized digital solution that encourages public participation and supports cleanliness initiatives towards building a cleaner and greener Goa.

CHAPTER 2

BRAINSTORMING AND PROJECT SELECTION

2. BRAINSTORMING AND PROJECT SELECTION

2.1 BRAINSTORMING

Brainstorming was the initial step in developing our project idea. Our team discussed various real-life problems and explored different possibilities for creating a software application that could be useful to society. During the discussion, we focused on issues related to cleanliness, waste management, and environmental protection in Goa. We aimed to develop a project that would encourage people to take responsibility for maintaining clean and healthy surroundings. Initially, we proposed an idea called Goa CoastCare, which focused mainly on maintaining cleanliness at beaches. However, after further discussion and evaluation, we decided to explore a broader approach to address cleanliness issues in different parts of Goa. This led to the development of our final project idea, CleanGoa, which connects citizens, volunteers, and administrators to report cleanliness issues and organize cleanliness drives.

2.2 INITIAL PROJECT IDEA: Goa CoastCare

The initial idea proposed by our team was Goa CoastCare, a project focused on improving and maintaining the cleanliness of beaches in Goa. Since Goa is well known for its beautiful beaches and tourism, keeping these coastal areas clean is important for protecting the environment and maintaining their natural beauty. During our brainstorming sessions, we discussed the problem of garbage and waste being scattered across beaches due to tourism and human activities. We wanted to develop a website that would help people report such cleanliness issues and encourage them to participate in keeping beaches clean.

The proposed website would allow users to report garbage by uploading photographs, providing a description of the issue, and specifying the location where the problem was found. These reports would help administrators identify areas that require attention and take appropriate action. The website would also provide opportunities for volunteers to participate in beach-cleaning drives and work together to maintain cleanliness. Users could receive information about upcoming cleaning activities and take part in protecting the coastal environment.

The main objective of Goa CoastCare was to create awareness about beach cleanliness, encourage responsible behaviour among citizens and tourists, and promote community participation in environmental protection. Through this project, we aimed to use technology to make reporting cleanliness issues easier and encourage people to contribute towards keeping Goa's beaches clean and attractive.

2.3 EVALUATION AND REJECTION OF INITIAL PROJECT IDEA

After proposing Goa CoastCare, the idea was evaluated to understand its usefulness, feasibility, and suitability for the project. Its main features, target users, and the problems it aimed to solve were considered. Although the idea focused on an important environmental issue, certain limitations were identified during the evaluation process.

The following points were considered during the evaluation:

- Limited Area of Application: The project was designed specifically for beach cleanliness and could not directly address garbage and waste problems in other public places, such as streets, markets, parks, and residential areas.
- Restricted Scope: Since the main focus was on beaches, the project had a limited range of cleanliness-related activities and reporting requirements.
- Limited Public Participation: The project mainly focused on coastal areas, restricting opportunities for people to participate in cleanliness activities across different parts of Goa.
- Practical Usefulness: The project needed to provide a platform that could address cleanliness problems in different locations and encourage people to maintain clean surroundings.

After looking at it closely, we decided not to go with Goa CoastCare. The main problem was that it only focused on cleaning beaches, which meant it couldn't be used for other garbage and trash issues in Goa. This helped us figure out what was missing and pushed us to find a better, more flexible idea.

2.4 FINAL PROJECT IDEA SELECTION: CleanGoa

After evaluating the initial idea, CleanGoa was selected as the final project to address cleanliness issues across Goa. The project focuses on developing a website where citizens can report garbage and waste problems in public places and participate in cleanliness activities. The website covers various locations, including beaches, streets, parks, markets, and residential areas. It aims to provide a common platform where cleanliness issues can be reported, reviewed, and monitored in an organized manner.

The website consists of two main modules: Cleanliness Reporting and Volunteer Mission Management. In the Cleanliness Reporting module, users can register and log in to their accounts, upload photographs of cleanliness issues, provide descriptions of the problems, and specify the location where the issue was found. Users can select a location on a map or enter the location details manually before submitting their reports. Administrators can review the submitted reports and approve or reject them based on the information provided. Approved reports are displayed on a public map, allowing users to view reported cleanliness issues in different areas and track their current status.

The Volunteer Mission Management module focuses on organizing and managing cleanliness drives. Administrators can create cleanliness missions by providing details such as the activity name, date, time, location, and description. Registered users can view upcoming missions, check the available details, and sign up to participate in the activities. This module helps coordinate volunteers and encourages people to work together to improve the cleanliness of their surroundings.

CleanGoa aims to connect citizens, volunteers, and administrators through a common platform for reporting cleanliness issues and coordinating community activities. It provides a systematic way to manage reports, monitor their progress, and organize volunteer participation. The website also aims to create awareness about waste management, environmental protection, and the importance of maintaining public spaces. Through these features, CleanGoa seeks to encourage responsible behaviour, improve community participation, and support efforts towards maintaining a cleaner and healthier environment across Goa.

2.5 REASONS FOR SELECTING CleanGoa

CleanGoa was chosen as our final project because it tackles real-world trash and cleanliness problems while getting the community involved in protecting the environment. It acts as a single, easy-to-use platform where everyday citizens, volunteers, and local authorities can all work together to report issues and plan cleanup activities.

The main reasons for selecting CleanGoa are:

- Wider Coverage: The project addresses cleanliness issues in different public places across Goa, including beaches, streets, parks, markets, and residential areas.
- Easy Issue Reporting: Users can report cleanliness problems by uploading photographs, providing descriptions, and specifying the location.
- Efficient Report Management: Administrators can review reports, approve or reject submissions, and update their status.
- Public Map Integration: The map allows users to view approved cleanliness reports and understand the status of issues in different locations.
- Volunteer Participation: The website enables administrators to organize cleanliness drives and allows users to register and participate.
- Environmental Awareness: The project encourages responsible behaviour and promotes the importance of maintaining clean surroundings.
- Practical Application: The project provides an opportunity to develop a useful website that combines issue reporting, location-based information, and volunteer management.

In the end, CleanGoa was chosen because it creates a simple, organized way for everyone to report trash problems and work together. By bringing citizens, volunteers, and local leaders onto one easy platform, it helps us team up to clean our neighborhoods, protect the environment, and build a much cleaner, healthier Goa for the future.

CHAPTER 3

TECHNOLOGIES USED

3. TECHNOLOGIES USED

3.1 HTML



HTML (HyperText Markup Language) is the standard markup language used to create and structure web pages. It provides the basic framework of a website by organizing content into elements such as headings, paragraphs, images, links, tables, forms, and buttons. HTML uses tags enclosed in angle brackets to define different elements of a web page. It works together with CSS and JavaScript to create functional and interactive websites. CSS is used to improve the appearance and layout of web pages, while JavaScript adds dynamic behaviour and interactivity.

In the CleanGoa project, HTML is used to develop the basic structure of the website and its various pages. It helps create user registration and login forms, cleanliness issue reporting forms, volunteer mission details, and navigation menus. HTML also provides elements for displaying photographs, descriptions, location information, and the status of reported cleanliness issues. Forms created using HTML allow users to enter the necessary information before submitting a report or registering for a cleanliness drive. Thus, HTML serves as the foundation of the CleanGoa website and supports the presentation of its main features in an organized and user-friendly manner.

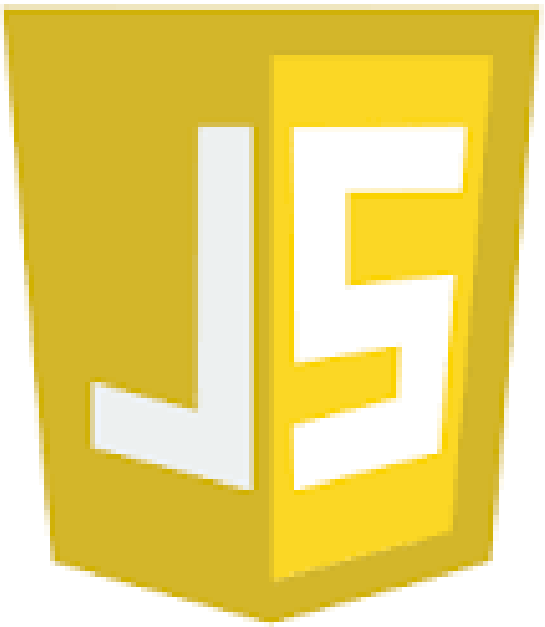
3.2 CSS



CSS (Cascading Style Sheets) is a style sheet language used to control the appearance and layout of web pages. It works with HTML to make websites visually attractive, organized, and user-friendly. CSS allows developers to set colours, fonts, backgrounds, margins, spacing, borders, and page layouts. It also supports responsive design, which helps websites adjust to different screen sizes, including computers, tablets, and mobile phones. By separating the styling from the HTML structure, CSS makes it easier to maintain a consistent design across multiple web pages.

In the CleanGoa project, CSS is used to improve the overall appearance and presentation of the website. It helps design the navigation bar, buttons, forms, report cards, and volunteer mission sections. CSS is also used to apply suitable colours, readable fonts, proper spacing, and organized layouts to different pages, including the login page, cleanliness reporting page, and mission management page. Additionally, CSS helps create a consistent interface so that users can navigate the website easily and access its features comfortably. Thus, CSS plays an important role in making the CleanGoa website attractive, responsive, and easy to use.

3.3 JAVASCRIPT



JavaScript is a programming language used to make websites interactive and dynamic. It works alongside HTML and CSS to provide functionality beyond the basic structure and appearance of a web page. JavaScript can respond to user actions, validate form inputs, display messages, update content dynamically, and handle events such as button clicks. It also supports communication with servers and databases through suitable APIs, allowing websites to process and display information without requiring a complete page reload.

In the CleanGoa project, JavaScript is used to add interactive features and improve the functionality of the website. It helps validate user inputs in registration, login, and cleanliness reporting forms before submission. It can also be used to handle photograph selection, display confirmation messages, and respond to user actions. JavaScript supports interactive map features, allowing users to select or view locations associated with cleanliness reports. It can also help display report statuses and volunteer mission details dynamically when connected to the appropriate backend services. Thus, JavaScript plays an important role in making the CleanGoa website interactive, responsive, and convenient for users.

3.4 JAVA



Java is a high-level, object-oriented programming language used to develop reliable, secure, and platform-independent applications. It follows the principle of Write Once, Run Anywhere (WORA), which means that Java programs can run on different operating systems using a compatible Java Virtual Machine (JVM). Java supports important object-oriented programming concepts such as classes, objects, encapsulation, inheritance, polymorphism, and abstraction. These concepts help organize code into reusable components, making software easier to develop, maintain, and manage.

In the CleanGoa project, Java is used as the primary programming language for implementing the application's backend logic and managing its main functionalities. It can be used to define classes for users, administrators, cleanliness reports, volunteer missions, and registrations. Java helps process submitted reports, manage user roles, validate information, update report statuses, and handle volunteer registrations. It can also connect the application to a database to store and retrieve user details, cleanliness reports, and mission information. By applying object-oriented programming principles, the CleanGoa project can maintain a structured and modular design. Thus, Java provides the programming foundation for implementing the core functionality of the CleanGoa system.

3.5 SPRINGBOOT



Spring Boot is an open-source Java framework used to develop web applications and backend services quickly and efficiently. It is built on the Spring Framework and simplifies application development by providing features such as automatic configuration, embedded servers, and ready-to-use dependencies. Spring Boot reduces the need for extensive manual configuration and helps developers create applications with a clear and organized structure. It also supports REST APIs, database connectivity, authentication, and integration with other technologies.

In the CleanGoa project, Spring Boot is used to develop the backend that manages communication between the website's user interface, application logic, and database. It can handle user registration and login, process cleanliness reports, manage report approval and rejection, and update report statuses. It also supports the creation and management of volunteer missions, including storing mission details and processing user registrations for cleanliness drives. Spring Boot can provide REST APIs through which the frontend communicates with the backend to send and retrieve information. Additionally, it can integrate with a database to store user details, report information, and mission records. Thus, Spring Boot helps organize and manage the core backend operations of the CleanGoa.

3.6 MYSQL COMMUNITY EDITION



MySQL Community Edition is a free, open-source relational database management system (RDBMS) used to store, organize, and manage data efficiently. It uses Structured Query Language (SQL) to create databases, tables, and relationships, as well as to insert, retrieve, update, and delete records. MySQL supports data integrity, primary keys, foreign keys, and constraints to maintain accurate and consistent information. It is widely used in web development because it can handle structured data and integrate with programming languages and backend frameworks such as Java and Spring Boot.

In the CleanGoa project, MySQL Community Edition is used to store and manage the information required by the website. It can maintain user registration details, cleanliness reports, descriptions, report locations, approval statuses, volunteer mission details, and participant registrations. The database allows the application to retrieve and update information whenever users submit reports, administrators review issues, or volunteers register for cleanliness drives. Tables can be connected through primary and foreign keys to maintain relationships between users, reports, missions, and registrations. Thus, MySQL Community Edition provides a structured and reliable way to manage the project's data and supports the smooth functioning of the CleanGoa website.

3.7 SPRING DATA JPA



Spring Data JPA is a part of the Spring ecosystem that simplifies database operations in Java applications. It provides an easy way to store, retrieve, update, and delete data from a relational database without writing repetitive SQL queries for every operation. It works with JPA, which defines how Java objects are mapped to database tables. Spring Data JPA provides repository interfaces and built-in methods that reduce the amount of code required for database access. It can also work with databases such as MySQL and integrates easily with Spring Boot applications.

In the CleanGoa project, Spring Data JPA is used to connect the backend application with the MySQL database and manage stored information. It helps handle data related to users, cleanliness reports, volunteer missions, and mission registrations. For example, Java classes can represent database entities such as User, CleanlinessReport, and VolunteerMission, while repository interfaces provide methods to save new reports, retrieve existing records, update information, and delete data when required. It also helps maintain relationships between different entities. By reducing repetitive database code, Spring Data JPA makes the CleanGoa application's data management more organized, efficient, and easier to maintain.

3.8 SPRING SECURITY



Spring Security is a Java framework that provides authentication and authorization features for web applications. It helps protect applications from unauthorized access by verifying user identities and controlling access to different features based on user roles and permissions. It supports secure login, password encryption, session management, and protection against common web security threats. Spring Security integrates with Spring Boot and can be configured to manage access to specific pages, backend services, and application resources.

In the CleanGoa project, Spring Security is used to protect user accounts and control access to the website's features. It helps manage the login process and ensures that users access only the functions permitted for their roles. For example, registered users can submit cleanliness reports, view approved reports, and register for volunteer missions, while administrators can review submitted reports, approve or reject them, update their status, and organize cleanliness drives. Passwords can be stored securely using password-hashing mechanisms rather than plain text. By providing authentication and role-based authorization, Spring Security helps protect user information and administrative functions, making the CleanGoa website safer and more reliable.

3.9 LEAFLET.JS



Leaflet.js is an open-source JavaScript library used to create interactive and mobile-friendly maps on websites. It allows developers to display maps, add markers, show locations, draw routes, and provide interactive map controls. Leaflet.js is lightweight, easy to use, and compatible with different map tile providers. It can be integrated with HTML, CSS, and JavaScript to display geographical information in a clear and user-friendly format. It also supports features such as zooming, panning, pop-up messages, and custom map markers.

In the CleanGoa project, Leaflet.js is used to display a public map showing cleanliness issues reported across Goa. It helps users identify the locations of reported garbage and waste problems through markers placed on the map. Users can click on these markers to view available information, such as the issue description and report status. The map can also help users select the location of a cleanliness issue when submitting a report. By displaying geographical information visually, Leaflet.js makes it easier to understand where cleanliness problems have been reported and helps users explore different locations. Thus, Leaflet.js supports the location-based features of the CleanGoa website and improves the accessibility of cleanliness information.

3.10 OPENSTREETMAP



OpenStreetMap (OSM) is a free, collaborative mapping platform that provides geographical data about roads, buildings, public places, and other locations around the world. It allows users and developers to access map data for different applications, including navigation, location tracking, and geographical information systems. OpenStreetMap data can be displayed on websites using mapping libraries such as Leaflet.js. It provides map tiles that show streets, landmarks, and other geographical features, helping users understand locations visually.

In the CleanGoa project, OpenStreetMap is used as the map data source for displaying geographical information across Goa. It works with Leaflet.js to present an interactive map where users can view the locations of approved cleanliness reports. Markers can indicate places where garbage and waste issues have been reported, allowing users to identify affected areas. The map can also assist users in selecting the location of a cleanliness issue while submitting a report. By using OpenStreetMap, CleanGoa can provide a clear visual representation of reported issues without requiring a proprietary map platform. Thus, OpenStreetMap supports the website's location-based reporting and public map features.

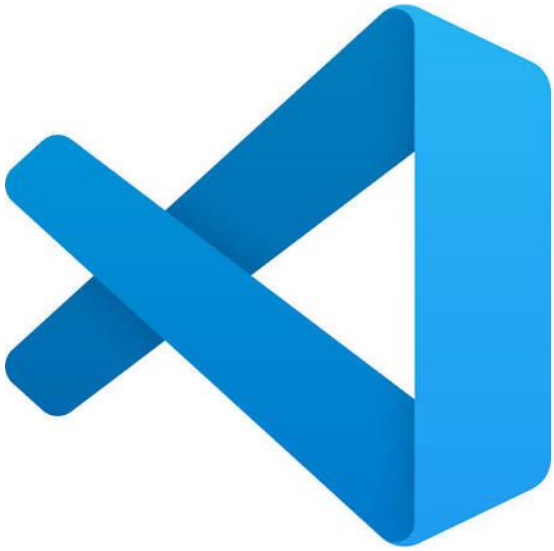
3.11 MAVEN



Apache Maven is a build automation and project management tool mainly used for Java applications. It helps developers organize their projects, manage required libraries, compile code, run tests, and prepare applications for deployment. Maven uses a configuration file called pom.xml (Project Object Model) to store project details, dependencies, plugins, and build settings. It automatically downloads the required libraries, so developers do not need to download and add each library manually. Maven also follows a standard folder structure, making it easier to organize and maintain project files. It provides simple commands to build and test applications, reducing manual work during development.

In the CleanGoa project, Apache Maven is used to manage the Java backend and its required libraries. It helps integrate technologies such as Spring Boot, Spring Data JPA, and Spring Security. The pom.xml file contains the dependencies required for backend development and MySQL database connectivity. Maven helps compile the Java code, run tests, and package the application into an executable JAR file when configured correctly. If additional libraries are needed, their dependencies can be added to the configuration file. This makes the project easier to develop, update, and maintain. Overall, Apache Maven simplifies the development process by managing dependencies and organizing the build activities of the CleanGoa application.

3.12 VS CODE



Visual Studio Code (VS Code) is a free and lightweight source code editor developed by Microsoft. It is used to write, edit, test, and manage code in different programming languages, including Java, JavaScript, HTML, and CSS. It provides useful features such as syntax highlighting, code suggestions, error detection, debugging, and extensions that make programming easier. VS Code also supports integration with tools such as Git and terminals, allowing developers to run commands and manage project files within the same application. Additional extensions can be installed to support different programming languages and frameworks.

In the CleanGoa project, VS Code is used as the main development environment for writing and managing the website's source code. It helps organize files related to the frontend, including HTML, CSS, and JavaScript, as well as Java files used for backend development. With suitable Java and Spring Boot extensions, VS Code can also help run and debug the backend application. The integrated terminal can be used to execute Maven commands, build the project, and run the application. Furthermore, VS Code allows developers to edit configuration files and manage the overall project structure efficiently. Thus, VS Code provides a convenient environment for developing, testing, and maintaining the CleanGoa website.

3.13 OOT CONCEPTS

Inheritance

Inheritance allows one class to acquire the properties and methods of another class.

Use in CleanGoa: Citizens and administrators share common features such as login and user details, while each has different functions. This reduces code repetition.

Polymorphism

Polymorphism allows the same method to perform different actions depending on the situation.

Use in CleanGoa: Citizens can view their reports and join cleanliness drives, while administrators can review reports and organize volunteer missions using different dashboard behaviours.

Abstraction

Abstraction hides complex implementation details and displays only essential features to users.

Use in CleanGoa: Citizens can report garbage by uploading photographs, entering descriptions, and selecting locations without needing to understand the website's internal processing.

Encapsulation

Encapsulation combines data and methods and controls access to the data.

Use in CleanGoa: Cleanliness report details and their status can be protected from direct modification, allowing information to be accessed or updated through controlled operations.

CHAPTER 4

SYSTEM ANALYSIS AND WORKFLOW

4. SYSTEM ANALYSIS AND WORKFLOW

4.1 WORKFLOW DISCRIPTION

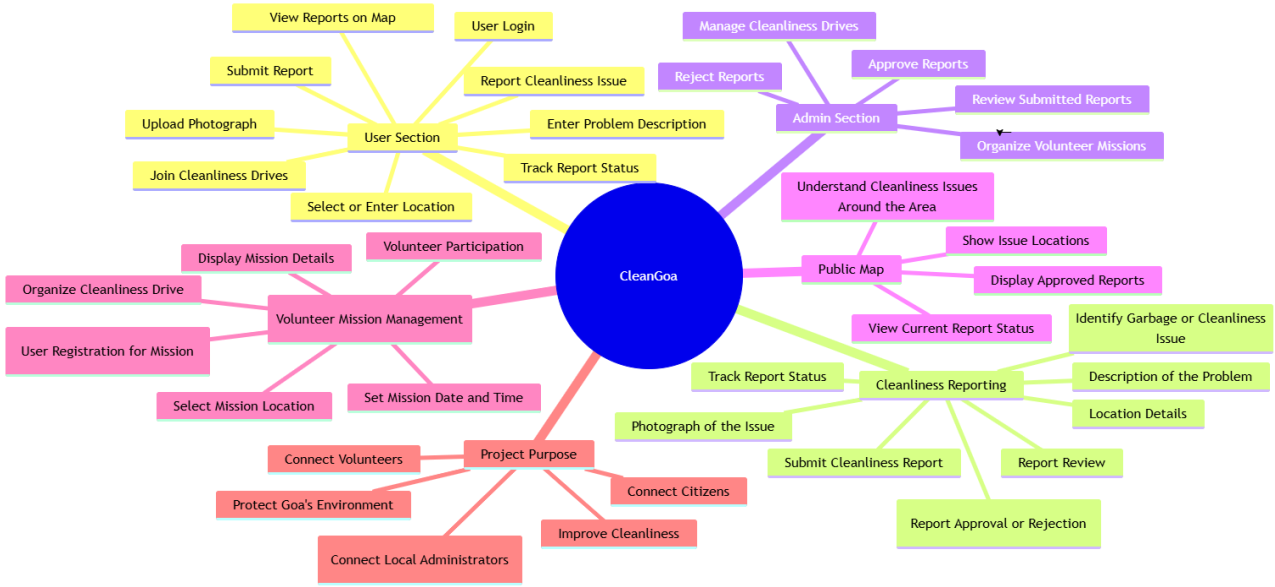
CleanGoa follows a systematic workflow to manage cleanliness issues and organize volunteer activities across Goa. The process consists of four main steps: initialization, reporting, administrative review, and volunteering.

- Initialization: Users register on the website and log in to access its features. The website provides two main modules: Cleanliness Reporting and Volunteer Mission Management.
- Cleanliness Reporting: Users report garbage and waste problems by uploading a photograph, entering a description, and specifying the location of the issue. After checking the details, they submit the report through the website.
- Administrative Review: The administrator reviews the submitted reports and either approves or rejects them. Approved reports are displayed on the public map, allowing users to view reported issues and track their status.
- Volunteering: The administrator organizes cleanliness drives by providing details such as the activity name, date, time, and location. Registered users can view upcoming missions, register for activities, and participate in cleanliness drives.

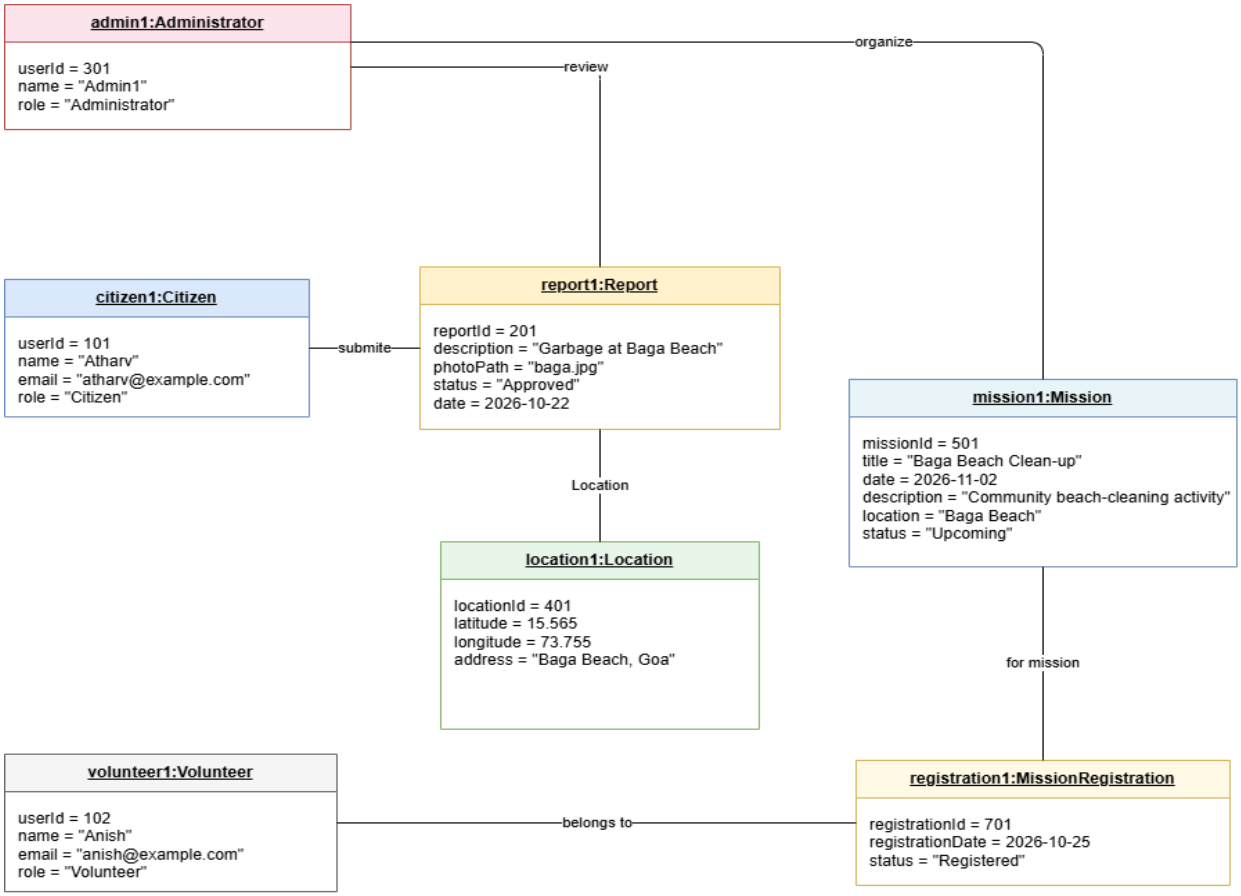
The website stores relevant information about users, reports, report statuses, and volunteer registrations. These details help manage cleanliness-related activities in an organized manner.

Overall, CleanGoa connects citizens, administrators, and volunteers through a common platform. It aims to simplify cleanliness reporting, improve coordination, encourage community participation, and support efforts to maintain a cleaner and healthier environment across Goa.

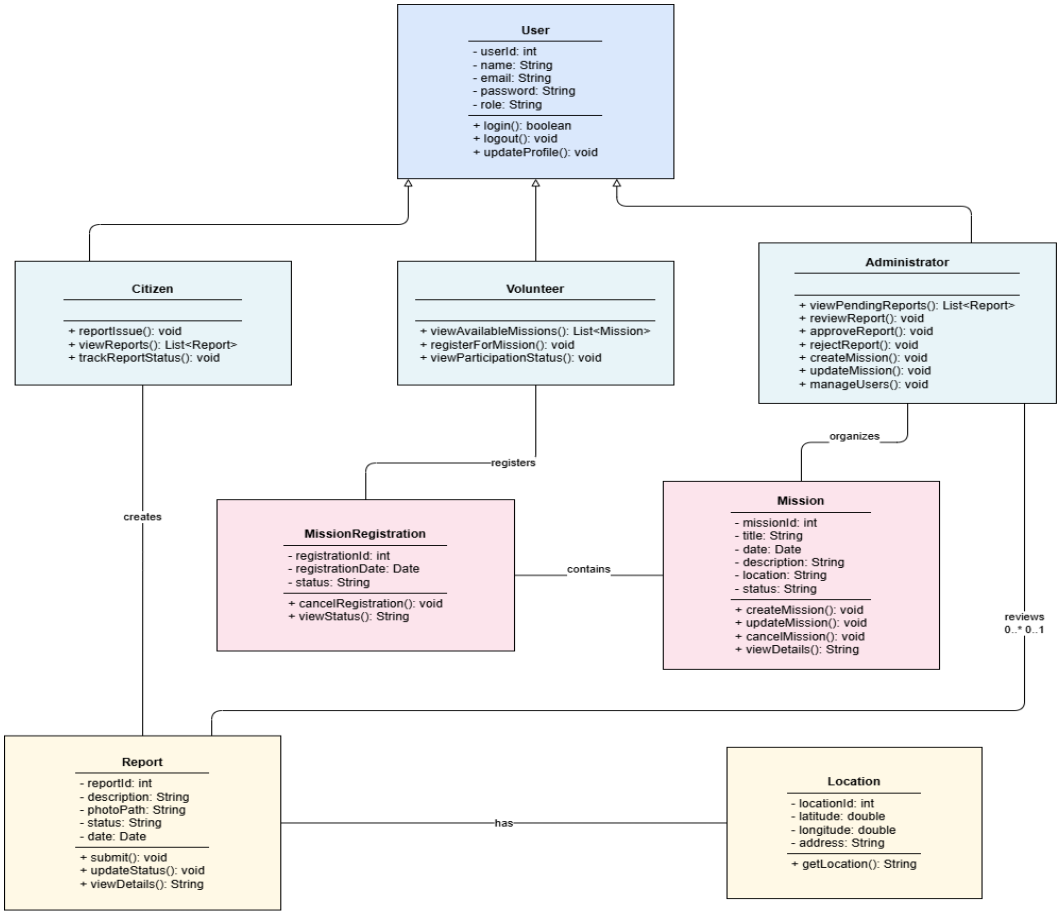
4.2 BRAINSTORMING INFO GRAPHIC DIAGRAM



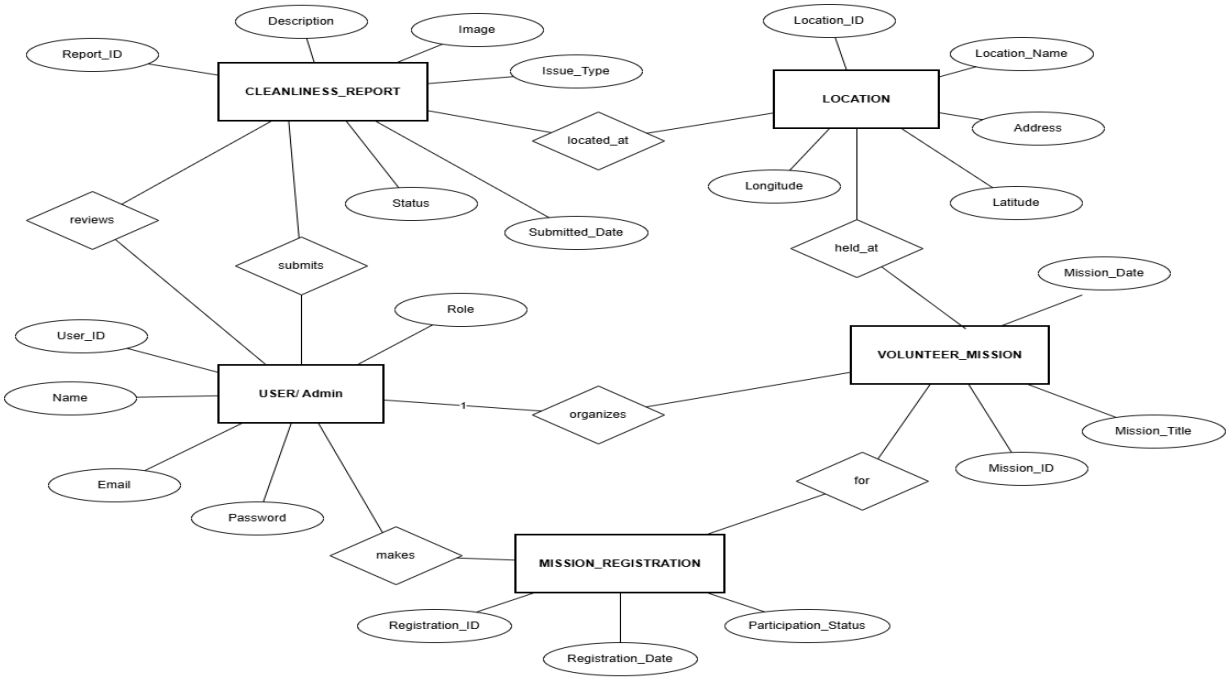
4.3 OBJECT DIAGRAM



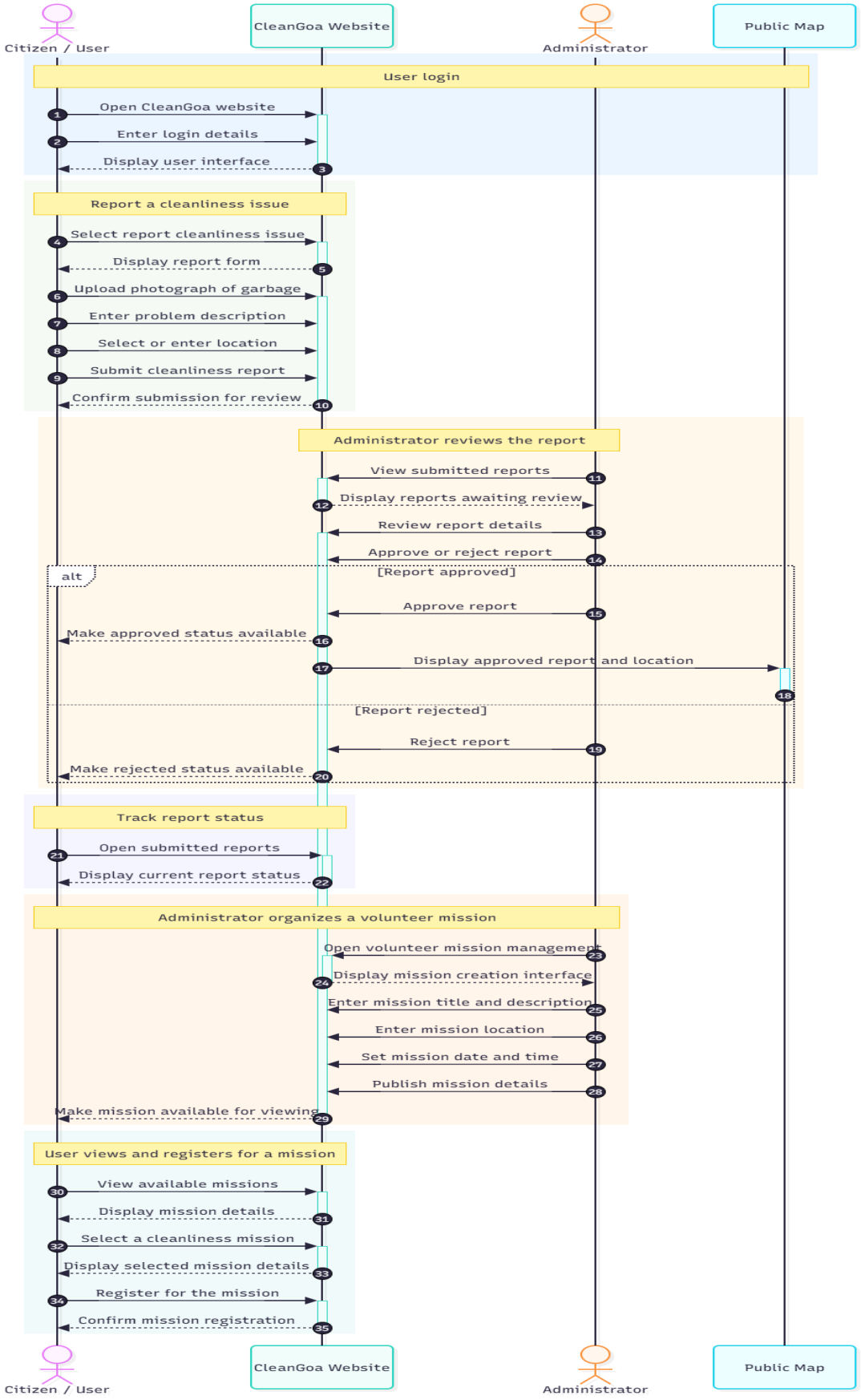
4.4 CLASS DIAGRAM



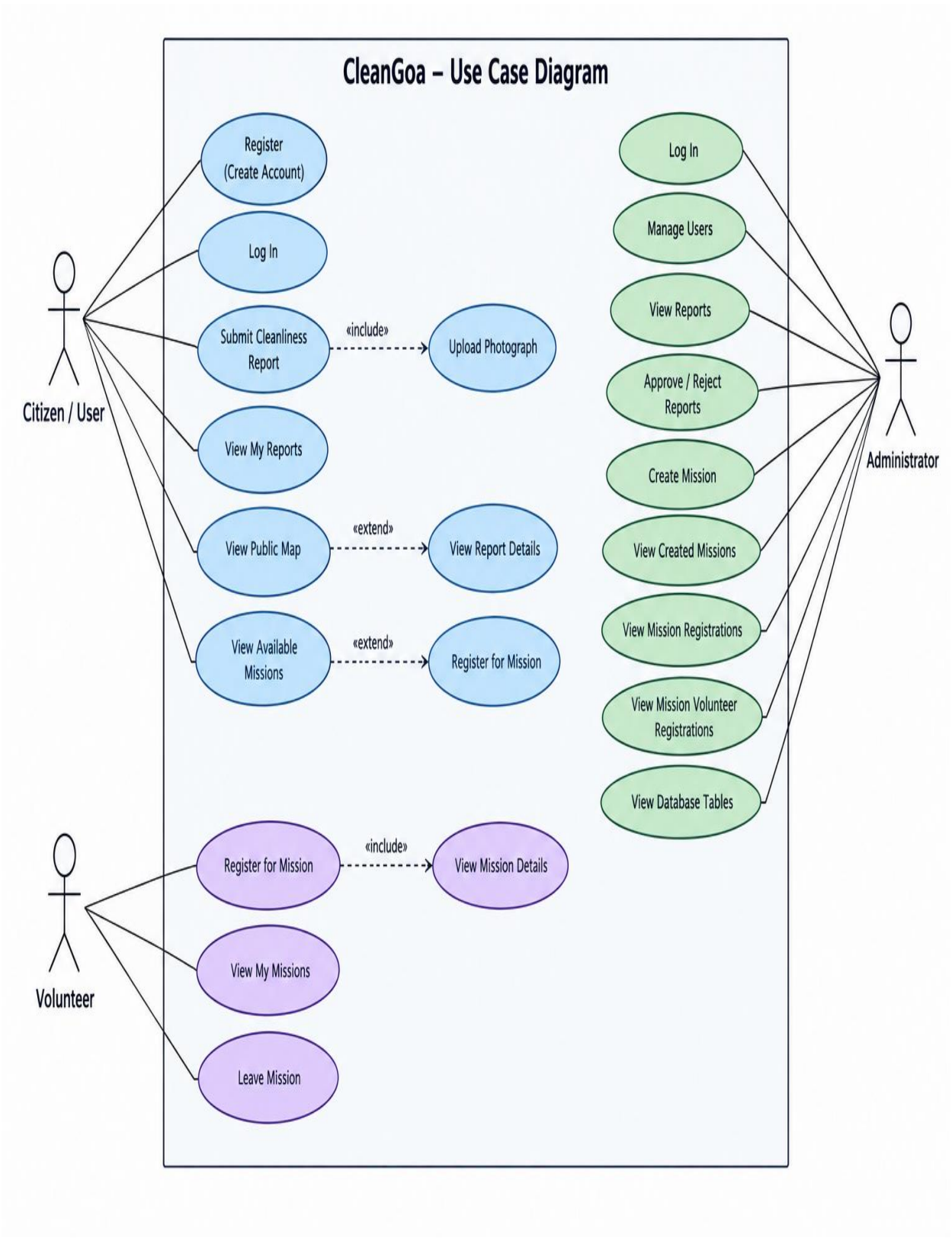
4.5 ENTITY RELATIONSHIP DIAGRAM



4.6 SEQUENCE DIAGRAM



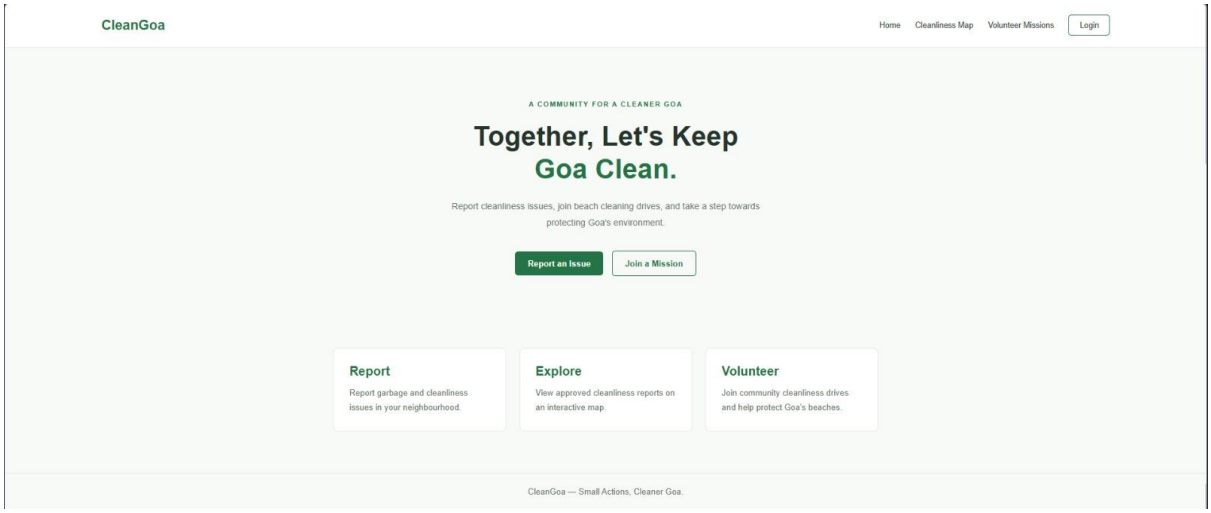
4.7 USE CASE DIAGRAM



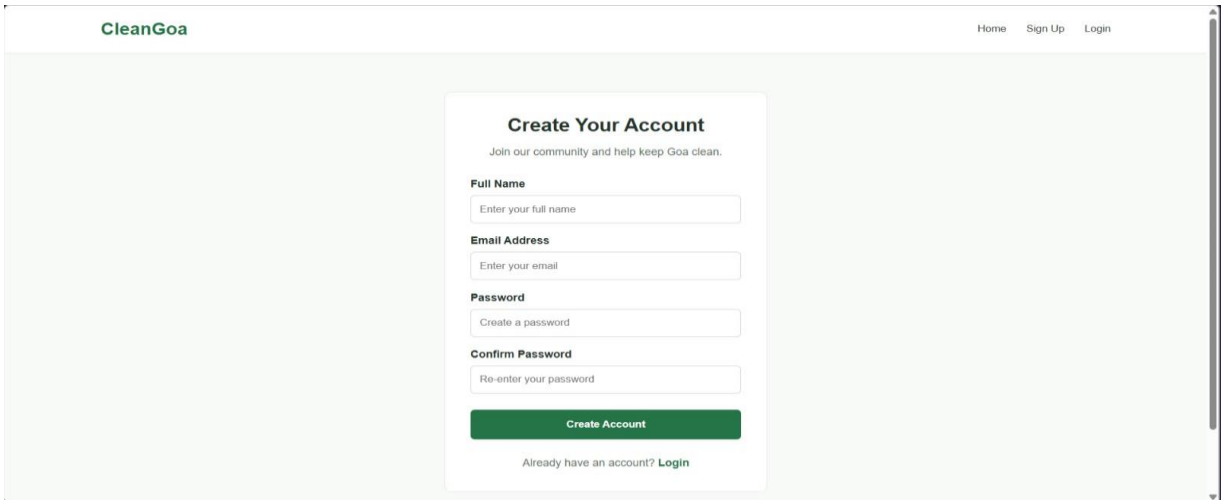
4.8 WORKING PRODUCT

4.8.1 USER MANUAL

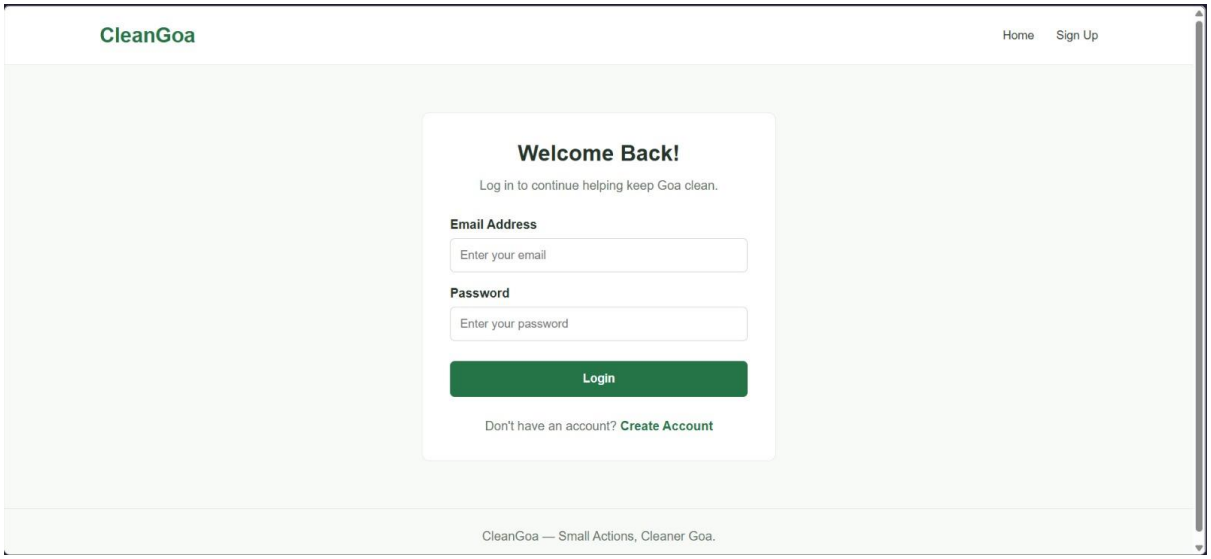
The main page of CleanGoa provides users with an overview of the website and its main features. It helps citizens access cleanliness reporting, view cleanliness issues on the public map, and participate in volunteer cleanliness drives.



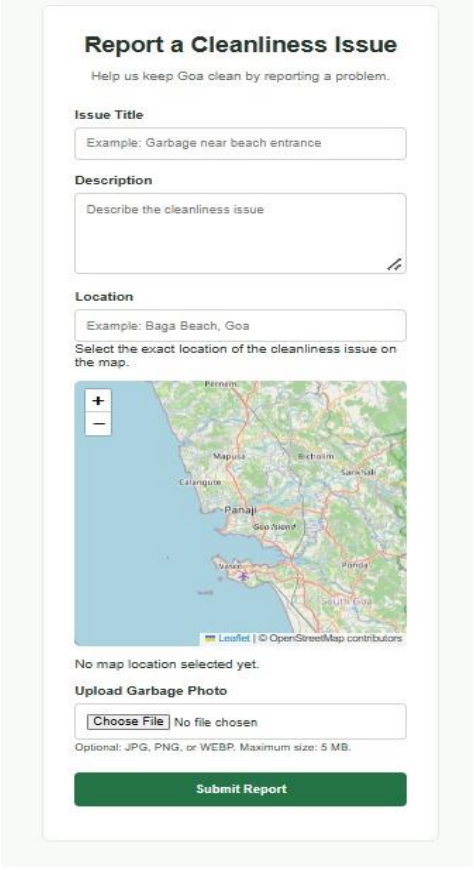
The main page is designed with a simple and user-friendly interface to make it easy for people to use the website and contribute to keeping Goa clean.



Users can navigate through the available options to report garbage, track the status of submitted reports, and view upcoming missions. The page also provides access to login and registration for users and administrators.

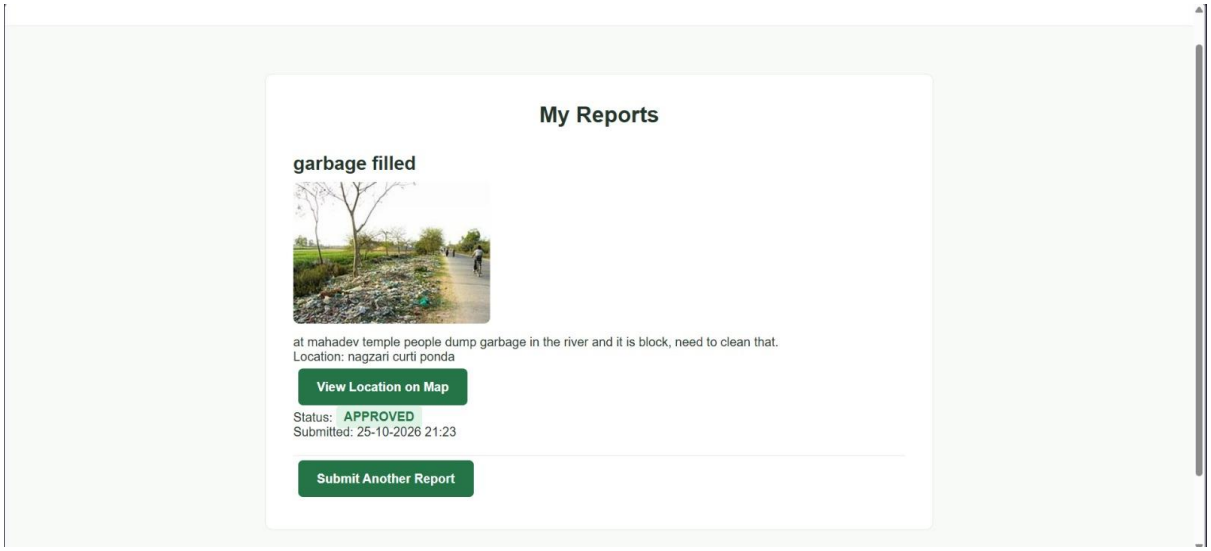


The Report Issue page allows citizens to report cleanliness problems in their surroundings.



Users can upload a photograph of the garbage, enter a description of the issue, and select the location on the map or enter it manually. After checking the details, users can submit the the report for administrative review.

The My Reports page allows users to view and track the cleanliness issues they have submitted through CleanGoa. Users can check the details of each report, including the issue description, reported location, and current status. The status helps users understand whether their report is pending review, approved or rejected.

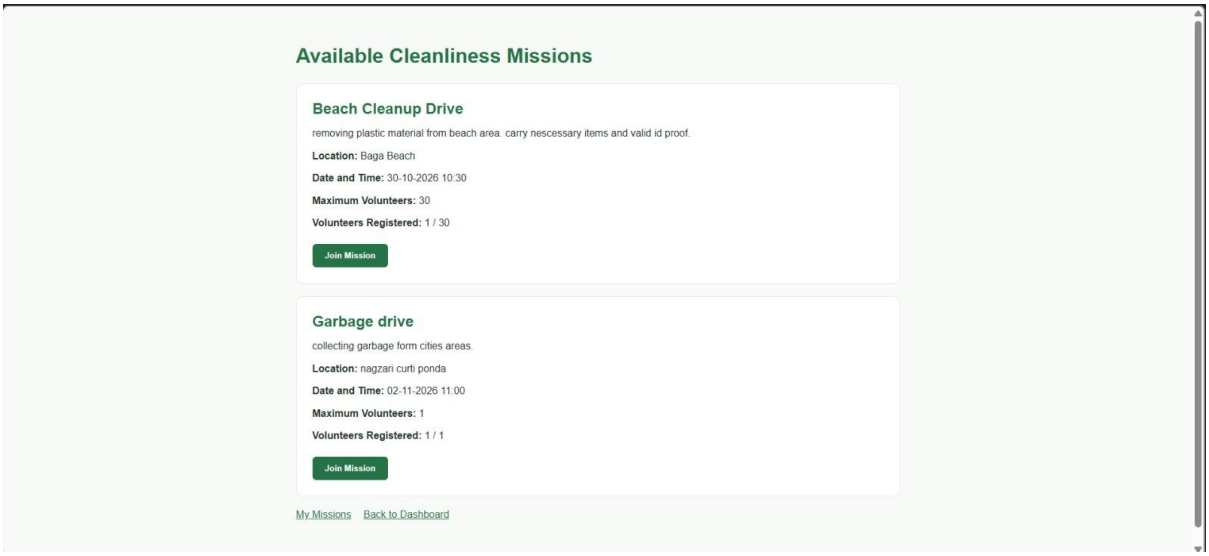


This page helps users stays informed about their submitted complaints and monitor the progress of cleanliness issues in their area.

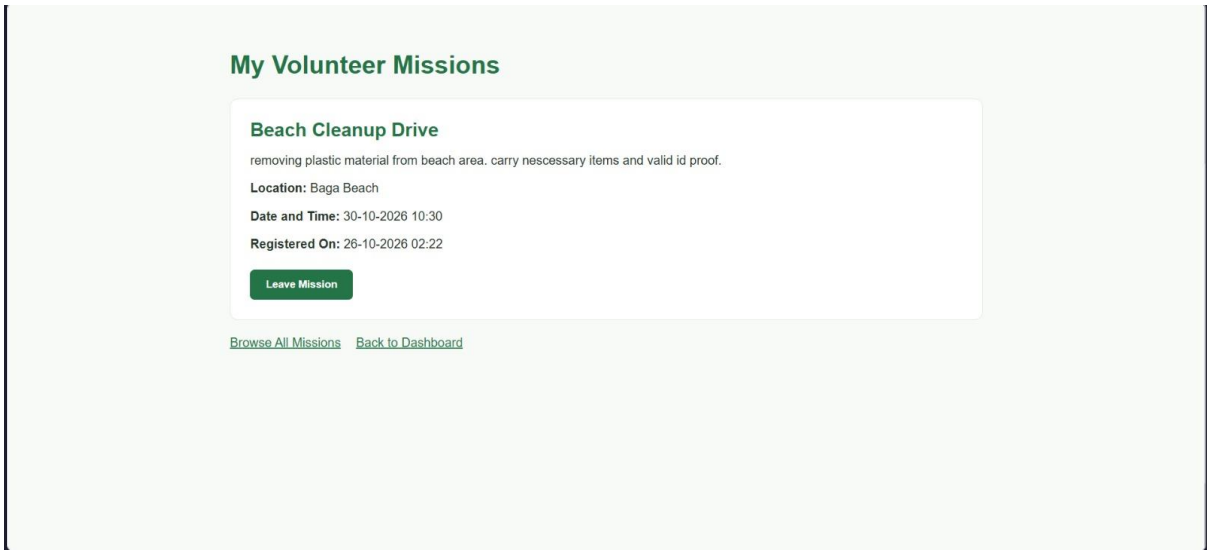
The Map and Missions section allows users to view reported cleanliness issues and participate in volunteer cleanliness drives. The map displays approved cleanliness reports along with their locations and current status helping users identify affected areas.



The Missions section provides details about organized cleanliness drives, including the location, date, time, and other relevant information. Users can view available missions, register for a drive, and participate in activities to help keep Goa clean. This section encourages community participation and supports environmental protection.



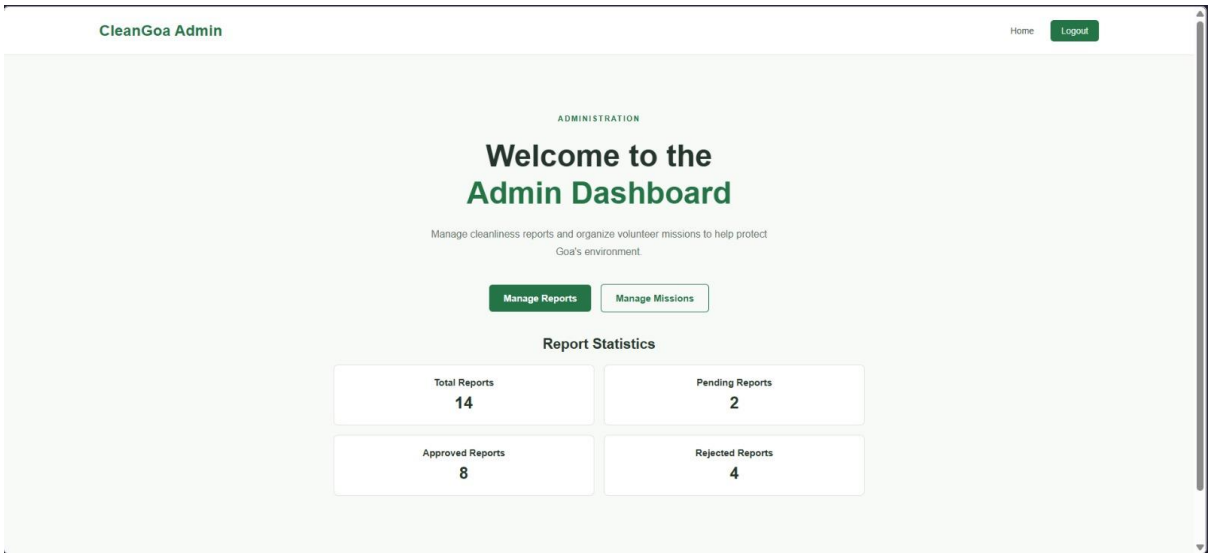
The My Volunteer Missions page allows users to view the cleanliness drives they have registered for through CleanGoa. Users can check important mission details, such as the mission name, location, date, and time.



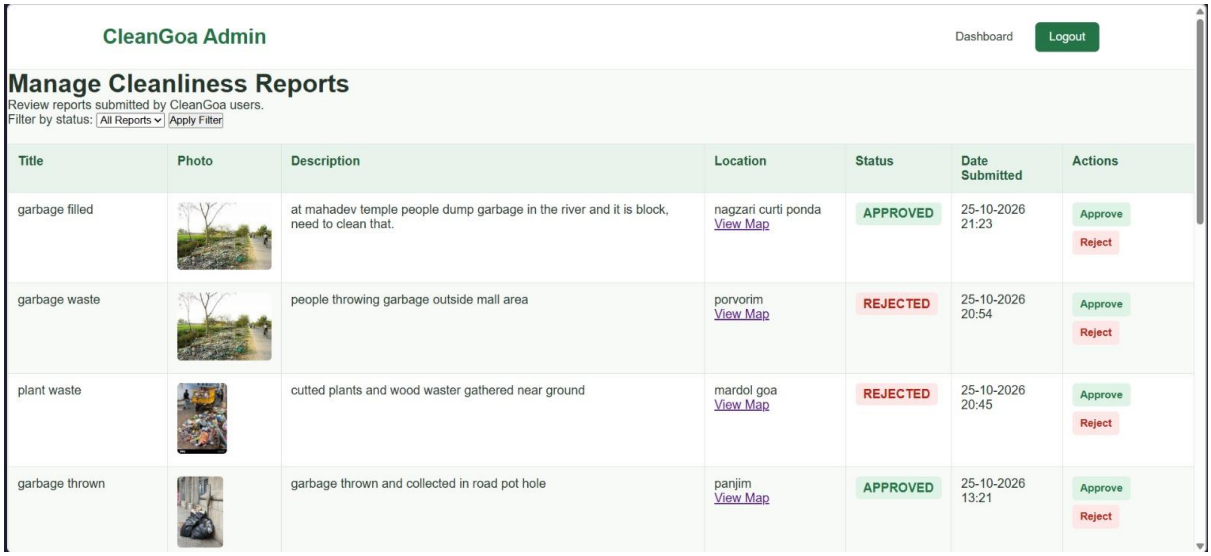
The Leave Mission option allows users to withdraw from a cleanliness drive they have previously joined. Users can select the mission they no longer wish to participate in and choose the Leave Mission option.

This page helps volunteers keep track of their upcoming cleanliness activities and know where and when to participate. It encourages users to stay involved in community cleaning activities and contribute to keeping Goa clean and environmentally friendly.

The Admin Dashboard allows the administrator to manage the main activities of the CleanGoa website.



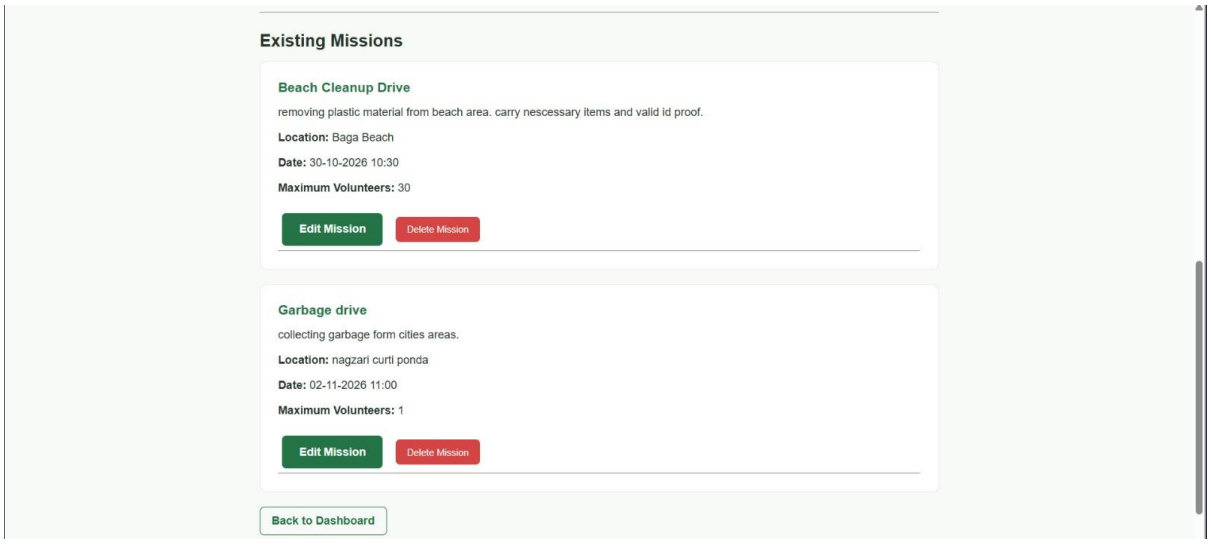
It provides access to submitted cleanliness reports, allowing the administrator to review, approve, or reject reports.



The administrator can also organize volunteer missions by adding mission details such as location, date, and time.



The Existing Missions page allows the administrator to view and manage cleanliness missions that have already been created in CleanGoa.



Through the dashboard, the administrator can monitor reported cleanliness issues and manage volunteer activities. This helps ensure that reports are properly reviewed and cleanliness drives are organized efficiently to support a cleaner Goa.

The Database Table section allows the administrator to view stored information related to the CleanGoa website. It contains records associated with users, cleanliness reports, and volunteer missions.

The Users Table displays the information of registered users on the CleanGoa website.

id	email	name	password	role
1	ajaynaik@gmail.com	ajay naik	$2a$10$J3KvN...	USER
2	a@gmail.com	anish	$2a$10$M/TJEj...	USER
3	admin@cleangoa.local	CleanGoa Administrator	$2a$10$tSAFM...	ADMIN
4	atharv@gmail.com	atharv naik	$2a$10$h8SBu...	USER
5	anishnyk@gmail.com	Anish Naik	$2a$10$/BzMihl...	USER
NULL	NULL	NULL	NULL	NULL

The Report Table displays the cleanliness issues submitted by users through the CleanGoa website.

id	created_at	description	image_path	latitude	location_name	longitude	status	title	user_id
1	2026-10-24 16:02:31.504335	Plastic waste and food wrappers are scattered ...	NULL	NULL	Baga Beach, Goa	NULL	APPROVED	Garbage near beach entrance	1
2	2026-10-24 16:11:38.797833	need to check the dustbin overflowing	NULL	NULL	panji bus stand	NULL	REJECTED	dustbin overflow	2
3	2026-10-24 16:28:12.090204	collected bottle garabage near beach	NULL	NULL	santa cruz	NULL	APPROVED	bottle garabage	1
4	2026-10-24 19:39:17.804457	garbage thrown at river near marcel	NULL	NULL	marcel goa	NULL	APPROVED	riverside garbage	4
5	2026-10-25 00:07:45.740507	garbage thrown at roadside	59f66d9f-89e8-412d-997c-2b1abbb6378a.jpg	NULL	panjim main circle	NULL	PENDING	garbage	1
6	2026-10-25 00:28:37.996350	dustbin overflow from past 3 days need urgent ...	NULL	NULL	talegoa	NULL	REJECTED	dustbin overflow	1
7	2026-10-25 00:57:17.183999	garbage collected near sports complex ponda.	d03eacc4-af37-442d-8788-a9bf5a85de5a.jpg	15.4066814992062	sports complex ponda	74.00512135252828	APPROVED	garbage near sports complex	1
8	2026-10-25 11:25:30.418123	garbage thrown near the gateway of my house	bb81f17f-cd8f-4bc9-b479-6281f5e93c21.jpg	15.41393408674527	farmagudi	73.98783954556929	APPROVED	garbage issue	4
9	2026-10-25 12:03:41.543841	plastic thrown near riverside	NULL	NULL	olg goa	NULL	PENDING	plastic thrown	4
10	2026-10-25 12:56:35.652241	local area garbage collected near light pole	79c04a3e-9efa-4bfc-b320-e7c9a7dd7856.jpg	15.40648290767423	mestawada curti	74.01868407597654	APPROVED	Garbage collection	2

The Mission Volunteer Registration Table displays the records of volunteers registered for different cleanliness missions.

id	registered_at	mission_id	user_id
3	2026-10-25 19:15:48.452522	1	1
4	2026-10-25 19:21:33.431017	2	1
5	2026-10-26 02:22:33.891451	1	5
NULL	NULL	NULL	NULL

The Created Mission Table displays the details of cleanliness missions created by the administrator on the CleanGoa website.

id	created_at	description	event_date	location_name	max_volunteers	title
1	2026-10-25 15:47:45.119368	removing plastic material from beach area. carr...	2026-10-30 10:30:00.000000	Baga Beach	30	Beach Cleanup Drive
2	2026-10-25 19:20:44.903101	collecting garbage form cities areas.	2026-11-02 11:00:00.000000	nagzari curti ponda	1	Garbage drive
NULL	NULL	NULL	NULL	NULL	NULL	NULL

4.8.2 SYSTEM TESTING

Test ID	Test Case	Expected Result	Status
TC01	Open the main page	Main page displays correctly	Pass
TC02	Register a new user	User registration is completed successfully	Pass
TC03	Log in to the website	User is redirected to the appropriate dashboard	Pass
TC04	Submit a cleanliness report	Report details and photograph are submitted	Pass
TC05	View My Reports	User's submitted reports are displayed	Pass
TC06	View the public map	Approved cleanliness reports appear on the map	Pass
TC07	View available missions	Existing cleanliness missions are displayed	Pass
TC08	Register for a mission	User is registered for the selected mission	Pass
TC09	Leave a mission	User's mission registration is cancelled	Pass
TC10	Review a cleanliness report	Admin can approve or reject the report	Pass
TC11	Create a new mission	Mission details are saved successfully	Pass
TC12	View the Users Table	Registered user records are displayed	Pass
TC13	View the Report Table	Cleanliness report records are displayed	Pass
TC14	View the Created Mission Table	Created mission records are displayed	Pass
TC15	View mission volunteer registrations	Registered volunteer records are displayed	Pass

CHAPTER 5

FUTURE SCOPE AND CONCLUSION

5.1 FUTURE SCOPE

The CleanGoa project can be improved in the future by adding more features to make cleanliness reporting and volunteer participation easier and more effective. The following features can be included in future versions of the website.

- Point-Based Reward System: A point-based system can be introduced to reward users for reporting genuine cleanliness issues and participating in cleanliness drives. Users can earn points based on their contributions, encouraging active participation in keeping Goa clean.

- Leaderboard System: A leaderboard can display users who have earned the highest points through their contributions. This can encourage healthy competition among users and motivate more people to participate in cleanliness activities.

- Notification System: A notification system can be added to inform users about report approvals or rejections, changes in report status, upcoming cleanliness drives, and volunteer mission updates.

- AI-Based Image Verification: Artificial Intelligence (AI) can be integrated to analyse uploaded photographs and identify whether they actually show a cleanliness issue. It can help detect garbage in images and flag unclear, irrelevant, or unsuitable photographs for further review by the administrator.

- Location-Based Notifications: The system can notify registered volunteers about cleanliness issues and upcoming drives near their location, making it easier for them to participate in nearby activities.

5.2 CONCLUSION

CleanGoa is a web-based project developed to address cleanliness issues across Goa by connecting citizens, administrators, and volunteers through a common platform. The main objective of the project is to make cleanliness reporting easier and encourage public participation in maintaining clean and healthy surroundings. The website consists of two main modules: Cleanliness Reporting and Volunteer Mission Management, which help manage cleanliness issues and organize community activities.

The Cleanliness Reporting module allows users to report garbage and waste problems by uploading photographs, providing descriptions, and specifying the location of the issue. Administrators can review the submitted reports, approve or reject them, and update their status. Approved reports are displayed on a public map, allowing users to view reported issues and track their progress. The Volunteer Mission Management module allows administrators to organize cleanliness drives by providing details such as the date, time, and location. Registered users can view upcoming missions and participate in these activities.

The project uses technologies such as HTML, CSS, and JavaScript for the frontend, while Java and Spring Boot are used for backend development. MySQL is used to store and manage the required information, with Spring Data JPA supporting database operations. Other technologies, such as Spring Security, Leaflet.js, and OpenStreetMap, support user authentication and location-based features.

Through this project, the importance of technology in addressing environmental problems can be understood. CleanGoa aims to improve coordination between citizens, administrators, and volunteers while promoting responsible waste management and environmental awareness. Overall, the project provides an organized approach to reporting cleanliness issues and encourages community involvement in maintaining a cleaner and healthier Goa.

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